But what about Flash memory and optical (holographic) storage and such? Will these not overtake hard drive storage at some point of time? At some point of time, yes, but not right now. Seagate claims that magnetic recording will remain the preferred form of mass storage. "With technologies (like holographic optical storage), what you've got is something that's much more expensive for the storage density you get," Seagate spokesperson John Paulsen said in 2002. "Magnetic technologies are mature, and they've been on a trajectory."

As for Flash memory, Jim Handy, director of non-volatile memory services at Semico Research writes, "Flash will be the leading choice in portable applications where a limited number of small files are used because it will offer the lowest overall system cost. Other limited-capacity applications will also gravitate towards Flash. In applications where file size or the total number of files to be stored is of more concern than total system cost, HDDs will prevail." There are further conclusions that Handy derives, but what we are to suppose is that there is still a long time before the hard disk goes the way of the floppy.

However, like we hinted at earlier, there will soon be a major shift in the way data is recorded on hard disk platters. Here, we'll be talking about AFC media, which could extend the life of LR a little while; perpendicular recording, or perpendicular magnetic recording (PMR); heat-assisted magnetic recording (HAMR, pronounced 'hammer'); and patterned media. But before all this, we need to talk a little about how a hard disk works.

Inside The Black Box

Hard disks have hard platters that hold the magnetic medium, as opposed to the flexible plastic film found in floppies. That's why they're called 'hard' disks. Each hard disk may have one or several platters. A platter is typically made either of aluminium alloy or a mixture of glass and ceramic. Both sides of the platters are coated with a magnetic medium.

The head—similar in function to the head in a tape recorder or VCR—flies over the disk, a fraction of a millimetre above it. Overall, the disk is a sealed box with controller electronics attached to one side. The electronics control the read/write mechanism as well as the motor that spins the platters. The electronics also assemble the magnetic domains on the drive into bytes for reading, and turn bytes into magnetic domains for writing. These electronics are contained on a small board that can be detached from the rest of the drive.

There's an arm that holds the read/write heads. It can move the heads from the hub to the edge of the disk. The arm and its movement mechanism are extremely light, fast, and precise.

Now comes the interesting part: how are the bits stored on the magnetic medium? The magnetisation is done on clusters of magnetic grains. About a hundred grains form a cluster. In LR, the magnetisation happens in the plane of the platter. Think of the clusters as little bar magnets placed in the same plane as the platter.

"HAMR looks very promising to us. Our view is that heat-assisted technology will be needed sooner rather than later"

Dr Mark Kryder
Director of Seagate Research and CTO of Seagate Technologies

According to Dr Mark Kryder, director of Seagate Research and CTO of Seagate Technologies, LR still has some time left before it is phased out altogether—it will take us beyond 100 gigabits psi. That's a very high areal density, but the world needs even higher densities. Areal density refers to how closely packed the magnetic clusters are. As areal densities increase, the superparamagnetic effect kicks in at some point.

Superpara-Wha?

The superparamagnetic effect, like we said, kicks in when the clusters that hold the bits get too closely packed together. To increase density, apart from packing the clusters closer together, they have to be made smaller, too. This is what is happening now in the hard disk world: we're just two product generations away from reaching the limit that superparamagnetism imposes on areal densities. You're unlikely to see a 2,000 GB hard disk manufactured using LR. (AFC media, like we said, could delay the death of LR for a while.)

So what exactly is the superparamagnetic effect? Superparamagnetism occurs when the magnetic grains on the disk become so tiny that random thermal vibrations at *room temperature* cause them to lose their ability to hold their magnetic orientations. What results are 'flipped bits'—bits whose magnetic north and south poles spontaneously reverse. This, naturally, corrupts the data, rendering it and the storage device unreliable and ultimately unusable.

If the problem lies in the bits getting demagnetised, why not just use stronger magnetisation? It turns out that the fields that can be applied are limited by the magnetic materials from which the head is made, and these limits are being approached, too!

Superparamagnetism is forcing the industry to slow the historically rapid pace of growth in drive capacity-a pace that, at its peak over the past decade, doubled capacity every 12 months. The way around superparamagnetism—or rather, the way to forestall its effects—is to use AFC media and/or perpendicular recording, scientists believe. This could create opportunities for continued growth in areal densities at a rate of about 40 per cent each year.

The exact areal density at which the superparamagnetic effect occurs has been subject to a lot of debate. Scientists in the 1970s predicted that the limit would be reached when data densities reached 25 megabits per square inch (psi)! But through constant innovation, such limits have been pushed forward by orders of magnitude.

In our discussion of how the limitations imposed by the superparamagnetic effect are being tackled, first up is AFC media.

AFC Media

AFC Media was designed to extend the life of LR. The idea is simple, but there are quite a few details under the hood. Regular media uses one layer of magnetic material; AFC media uses two layers, the top one thicker than the bottom one, with a three-atom-thick layer of ruthenium (a non-magnetic material) sandwiched in between. (IBM calls the ruthenium layer 'pixie dust'.) This